

Seminar I
Jožef Stefan International Postgraduate School
Ljubljana, Slovenia

Supervisor: Prof. Dr. Nada Lavrač, Jožef Stefan International Postgraduate School,
Ljubljana, Slovenia

Co-Supervisor: Prof. Dr. Matthew Purver, School of Electronic Engineering and Com-
puter Science, Queen Mary University of London, London, UK

MEDNARODNA PODIPLOMSKA ŠOLA JOŽEFA STEFANA
JOŽEF STEFAN INTERNATIONAL POSTGRADUATE SCHOOL



Nikola Ivačič

CROSS-LINGUAL NLP TECHNIQUES FOR SLOVENIAN NEWS ANALYSIS

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Supervisor: Prof. Dr. Nada Lavrač

Co-Supervisor: Prof. Dr. Matthew Purver

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Abstract

The ever-increasing amount of global news presents an overwhelming challenge for individuals and organizations striving to keep abreast of relevant information. The sheer volume of unstructured text data makes it nearly impossible for users to consume and comprehend all the information pertinent to their interests. Media monitoring and analysis systems can help us by employing advanced algorithms to address the problem of size and the problems of opinionated views and misinformation. These tools allow users to quickly glean the essential information from an article, saving time and reducing cognitive load. This paper presents an overview of two specific topics concerning news analysis and Natural Language Processing (NLP): Named Entity Recognition and News Framing. In the context of these NLP tasks, we recognize the potential for utilizing cross-lingual NLP techniques to augment the analysis of news in languages that lack sufficient annotated data. The primary goal of cross-lingual NLP is to transfer knowledge from high-resource languages to low-resource languages where annotated data is scarce or unavailable. We mainly focus on low-resource Slovene and other South Slavic languages with even less annotated data. Therefore we analyze key techniques in the needed cross-lingual setting, which include multilingual pre-training, transfer learning, and zero-shot learning.

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Chapter 1

Introduction

In this section, we briefly explore the significance of cross-lingual news analysis in the Slovene language for the Natural Language Processing (NLP) field and provide the motivation behind this paper.

1.1 News Monitoring and Analysis

Initially, the purpose of News monitoring and analysis was to relieve us from monitoring all published news; however, as the volume of news increased, it had to evolve and employ more advanced Natural Language Processing (NLP) techniques. News Monitoring and analysis can positively influence society because its objective is to present and analyse all news content impartially without favouring any particular viewpoint. Additionally, it benefits society by striving to identify biased perspectives, misinformation and sentiment within news content. Nowadays, media monitoring and analysis must be conducted in real-time, which involves organising articles based on their content, attaching various metadata categories, and synthesising information from multiple news sources. Natural Language Processing (NLP) techniques can effectively address these challenges by facilitating the development of personalised news recommendation systems, automated content summarisation, organising and prioritising news items, and ensuring users access the most pertinent and up-to-date information. These tools allow users to quickly glean the essential information from an article, saving time and reducing cognitive load.

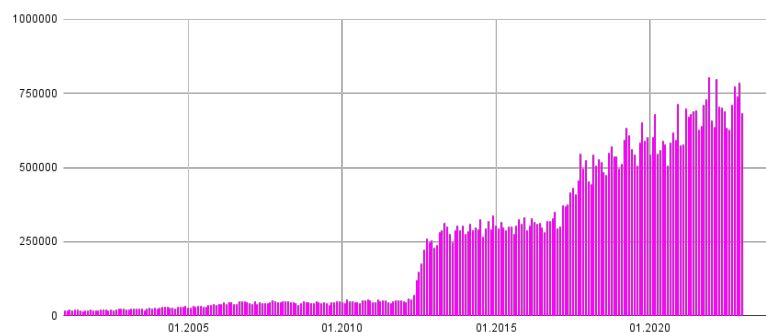


Figure 1.1: Increasing number of client-relevant news articles in the Balkan region monitored by the industry.

1.2 NLP, Slovene and Other South Slavic Languages

Creating solutions for under-resourced languages is crucial in increasing the presence of non-dominant languages within the digital realm. The western Balkan region, formerly also a common political entity, has common cultural and language origins. This allows us to explore also the possibilities of transfer learning in NLP between languages from the same family where training resources are scarce. The accessibility of annotated training materials for the Slovene language has experienced significant enhancement, thanks to the efforts of Common Language Resources and Technology Infrastructure, Slovenia (“Slovenian Language Resource Repository,” 2022) and The Jožef Stefan Institute (“Natural Language Server,” 2022). However, the size of training corpora for other South Slavic languages remains limited or, in some cases, absent. Additionally, Slovene and other South Slavic languages pose unique challenges in NLP since they are highly inflected, with a rich system of inflectional morphology for nouns, pronouns, adjectives, and verbs.

1.3 Cross-Lingual NLP Techniques

The main objective of cross-lingual NLP is to transfer knowledge from high-resource languages to low-resource languages where training data size is small or nonexistent. Fundamental techniques in the cross-lingual setting include multilingual pre-training, transfer learning, and zero-shot learning.

1.3.1 Large Pretrained Multilingual Models

Within Natural Language Processing, Transformer-based (Vaswani et al., 2017) deep neural networks have recently emerged as the prevailing approach. Large pre-trained multilingual language models (LPMLMs), such as mBERT (Devlin et al., 2018) and XLM-RoBERTa (Conneau et al., 2019), can solve nearly all linguistic tasks (e.g., information extraction, text classification, sentiment analysis, topic detection, etc.) better compared to conventional NLP techniques. LPMLMs are initially trained on unsupervised, broad corpora but can be fine-tuned for particular tasks, such as sequence or token classification.

The language model learned in pre-training reduces the amount of labelled data needed for fine-tuning. This is more than evident in the recent advancements of the text-to-text transformers like T5 (Raffel et al., 2020), Generative Pretrained Transformer (GPT) family (Brown et al., 2020), PaLM (Chowdhery et al., 2022), and LLama (Touvron et al., 2023), which are fine-tuned on a wide range of text-based tasks, including question-answering, text generation, and language translation to name a few.

Various experiments also achieved state-of-the-art performance in specific NLP tasks for the Slovene language. For instance, sentiment analysis (Pelicon et al., 2020), keyword extraction (Martinc et al., 2021), named entity recognition (Prelevikj & Žitnik, 2021), cross-lingual hate-speech detection (Pelicon et al., 2021), etc. All state-of-the-art NLP techniques presented in this paper use Transformer-based neural network architecture as its foundation.

1.3.2 Zero-shot and Few-shot Learning

The Zero-shot technique, first used in classification tasks with a target to predict new unseen classes (Chang et al., 2008; Larochelle et al., 2008; Palatucci et al., 2009), has broadened its meaning by use in various NLP tasks. The Zero-shot term (and technique) is also used as a cross-lingual model transfer in which task-specific annotations in one language are used to fine-tune the model for evaluation in another language (Pires et al.,

2019). Zero-shot cross-lingual model transfer use was also demonstrated from Slovene to Croatian language (Pelicon et al., 2020).

Until recently, task-specific fine-tuning was a prevailing method of achieving state-of-the-art results in NLP. GPT models have shown unprecedented performance in Zero-shot and Few-shot Learning, as was shown by Brown et al. (2020). Active research is being conducted on different architectures and pre-training objectives used in state-of-the-art models. Wang et al. (2022) demonstrated that the masked language objective with encoder-decoder architecture works best for Zero-shot NLP learning after multi-task fine-tuning. In contrast, Touvron et al. (2023) showed that the text-to-text transformer model can be much smaller (than GPT) and still competitive.

1.4 Evaluation Metrics

This section discusses quantitative evaluation measures for Named Entity Recognition (NER) and Computational Framing Analysis (CFA). Traditionally, NER is modelled as a multi-class classification problem, whereas CFA has recently been more commonly viewed as a multi-label classification problem. Therefore macro-averaged F1-score over all classes is used for NER, while for CFA, micro-averaged F1-score and Hamming Loss over all labels are also often used (Yang et al., 2018). Macro-averaged F1-score is commonly used to evaluate CFA model performance when exactly one frame is to be predicted.

1.4.1 F-measure

Traditionally, an F-measure or balanced F-score is used to evaluate the classification models. F-score is the harmonic mean of precision and recall and can have values between 0 and 1 (closer to 1 means a better score):

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Precision and Recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN}$$

given that:

- False Positives (FP): a class or label predicted but not present in the test.
- False Negatives (FN): a class or label is present in the test but missing in our prediction.
- True Positives (TP): a class or label that is correctly predicted.
- True Negatives (TN): a class or label that is correctly not predicted and not present in the test.

1.4.2 Macro F-scores

Macro-averaged F1-score is computed using the arithmetic mean of all the per-class $F1_i$ scores:

$$\text{Macro-averaged F1-score} = \frac{1}{n} \sum_{i=1}^n F1_i$$

where $F1_i$ is the F1-score for i th label or class.

1.4.3 Micro F-scores

Micro-averaging calculates a global average F1 score by aggregating the counts of True Positives (TP_a), False Negatives (FN_a), and False Positives (FP_a) for all classes or labels. Micro-averaged F1-score can then be derived from Precision and Recall equations:

$$\text{Micro-averaged F1-score} = \frac{TP_a}{TP_a + \frac{1}{2}(FP_a + FN_a)}$$

1.4.4 Accuracy

The accuracy of the classification model is commonly represented as a fraction of correctly classified predictions overall predictions. In multi-class classification cases where each observation has a single label, the micro-F1 and accuracy share the same value, but in general, we can define accuracy as:

$$\text{Accuracy} = \frac{TP_a + TN_a}{TP_a + FP_a + FN_a + TN_a}$$

1.4.5 Hamming-loss

Hamming loss is a common multi-label evaluation metric defined as the fraction of the wrong labels to the total number of labels.

$$\text{Hamming loss} = \frac{1}{n \cdot l} \sum_{i=1}^n \sum_{j=1}^l \text{xor}(y_i, \hat{y}_i)$$

where n is the number of samples, l is the number of labels, y_i is the target, $\hat{y}_i$ is the prediction, and xor is exclusive or operator that returns zero when the target and prediction are identical and one otherwise.

Table 1.1: Example Hamming loss calculation: $\frac{5}{4 \cdot 5} = 0.25$ where each row contains a target y_i followed by a prediction $\hat{y}_i$ vector with five label values with five wrong predictions in total

	y_i	$\hat{y}_i$
x_1	[1, 0, 1, 1, 0]	[1, 0, 1, 0 , 1]
x_2	[0, 1, 1, 0, 0]	[0, 1, 1, 1 , 1]
x_3	[0, 0, 0, 1, 0]	[0, 0, 1 , 1, 1]
x_4	[1, 1, 0, 1, 0]	[1, 1, 0, 1, 0]

Contrary to F-score, a Hamming loss closer to 0 means a better score.

1.5 Scope and Objectives

The scope of this paper is to present two common NLP tasks and their state-of-the-art, Named Entity Recognition (NER) and Computational Framing Analysis (CFA). NER has a long history and is one of the most common NLP tasks that can be used for news analysis. CFA gained traction in NLP in recent years, where more and more emphasis is given to the analysis of implicit suggestion or persuasion in text. Both topics will be presented in the context of news analysis with a strong emphasis on the Slovene and other Slavic languages, where cross-lingual techniques are sometimes the only option to tackle the given problem.

For the CFA, we consider the state-of-the-art in other languages since we conducted only preliminary research in Zero-shot learning for the Slovene language.

1.6 Organization of the Paper

In the following chapter, we introduce NER, including its annotation methods and a concise history of NER research. We then explore the latest advancements in NER for the Slovene language and conclude with an overview of the training datasets.

In Chapter 3, we first present the CFA and its research history. Next, we discuss the current state-of-the-art in CFA. Finally, we conclude the chapter by exploring the available datasets and briefly outlining our initial research findings.

In conclusion, we critically analyse both NER and CFA topics, followed by a discussion of potential enhancements and an outline of our planned research endeavours for the future.

Chapter 2

Named Entity Recognition

Named Entity Recognition (NER) is one of the cornerstones of the NLP tasks and is widely used in many real-life applications, including in the news industry. Named entities, like persons (PER), places (LOC), organizations (ORG), and others (MISC), provide a context to the news story. They often directly answer some of the “Five Ws” questions (Who, What, When, Where, Why) that are considered basic questions of information gathering and journalism.

In the scope of Machine Learning, NER is usually considered a multi-class token classification problem, where each token in the text belongs to a specific NER class. Persons, locations, organizations, and others, which are considered traditional labels used in news corpora and most comparison benchmarks, were first presented in CoNLL-2003 shared task (Tjong Kim Sang & De Meulder, 2003). In various corpora, it is not unusual to find additional labels, such as labels for events and products (Piskorski et al., 2017), while some corpora possess highly intricate or even hierarchical NER labelling systems (Ševčíková et al., 2007).

2.1 Traditional NER Approaches

The history of NER research dates back to the 1990s, with initial studies conducted by researchers such as Grishman and Sundheim (1996), and followed by Sang and De Meulder (2003) and Segura-Bedmar et al. (2013), among others. During this period, the first rule-based models were developed (Yu et al., 2020). Those models were based on pre-defined patterns and hand-crafted rules (e.g., LTG, NetOwl) and could include regular expressions, gazetteers (lists of known named entities), and morphological patterns. These approaches were followed by the unsupervised methods (Collins & Singer, 1999; Nadeau et al., 2006), where no annotated data were required.

The advent of machine learning algorithms opened a novel direction for NER tasks. Various classification techniques, such as Support Vector Machines (SVM), Decision Trees, or Maximum Entropy classifiers, were used along with feature engineering, which used word information like capitalisation patterns, word shape, Part-Of-Speech tagging (POS), chunking, prefixes, suffixes, and even dictionaries (Florian et al., 2003; Krishnan & Manning, 2006).

Later, statistical, probabilistic graph models like Conditional Markov Models (Borthwick, 1999; Leek, 1997; McCallum et al., 2000) (CMMs) and Conditional Random Fields (CRFs), introduced by Lafferty et al. (2001) were applied to address the NER problem (Finkel et al., 2005). CMMs are generative, while CRFs are graphical probabilistic models that can be used for such sequence labelling tasks. CRFs model the conditional probability of the label sequence given the input sequence while considering the dependen-

cies between adjacent labels. CRFs were considered state-of-the-art for NER tasks before the emergence of neural network-based techniques. These models were successfully used in the Slovene language projects (Ljubešić et al., 2019) and research studies (Stajner et al., 2013).

2.2 Neural Network Approaches

One of the early neural network-based approaches for NER was introduced by Collobert et al. (2011). They proposed a Convolutional Neural Network (CNN) and a deep learning architecture called SENNA (Semantic/Syntactic Extraction using a Neural Network Architecture) for several NLP tasks: POS, chunking, NER and Semantic Role Labelling. The SENNA model used a lookup table for pre-computed word embedding and a sliding window approach with CNN to capture the local context around each word.

Subsequently, RNN-based models became popular for NER tasks. For example, Lample et al. (2016) proposed a bidirectional LSTM (Graves et al., 2013; Hochreiter & Schmidhuber, 1997) (Bi-LSTM) model with a Conditional Random Field (CRF) layer on top for NER. This approach combined the strengths of both sequence labelling (using Bi-LSTM) and structured prediction (using CRF) to achieve state-of-the-art performance at the time. An example implementation can be found in Stanza library (Qi et al., 2020) and CLASSLA derivative for Slovene language (Ljubešić & Dobrovoljc, 2019).

Later, it was shown that the multilingual BERT (mBERT) transformer model outperforms the Bi-LSTM-CRF model for the NER task, and the performance can be even further improved with a word-level CRF layer (Arkhipov et al., 2019).

2.3 Multilingual Models

Besides rich-resourced languages (e.g. English), there is a shift to several less-resourced ones, including the Slavic family (see several organized shared tasks (Piskorski et al., 2021; Piskorski et al., 2019; Piskorski et al., 2017)). Large pre-trained multilingual models (LPMLMs) have been used in a large number of tasks, including cross-lingual hate-speech detection Pelicon et al. (2021), zero-shot sentiment analysis (Pelicon et al., 2020) as well as for NER (Arkhipov et al., 2019; Suppa & Jariabka, 2021).

Nevertheless, it is also evident that XLM-RoBERTa outperforms BERT (Suppa & Jariabka, 2021) in such tasks. Prelevikj and Žitnik (2021) showed that the monolingual NER model performance for the Slovene language is practically equal to that of a multilingual one and has achieved state-of-the-art performance on the NER task.

2.4 Datasets and Resources for South-Slavic NER

NER corpora are usually presented as an additional annotation in larger corpora like dependency treebanks. For such corpora, the CoNLL format is the most widely used. In the CoNLL format, a text corpus is presented with sentences separated by a blank line, and each line contains a single word token with its attributes separated by a tab character (see Listing 2.1).

Listing 2.1: Example original CoNLL-2003 format

```
U.N. NNP I-NP I-ORG
official NN I-NP O
Ekeus NNP I-NP I-PER
heads VBZ I-VP O
for IN I-PP O
Baghdad NNP I-NP I-LOC
. . . O O
```

CoNLL2003 format was further developed to CoNLL-U, which supports additional word (token) level annotations like POS, lemmas, and dependencies, to name a few (de Marneffe et al., 2021). NER is usually annotated in IOB2 (Ramshaw & Marcus, 1995) format, which uses B and I prefixes for NER labels to mark the beginning and the inside of NER annotated text spans.

Listing 2.2: Example CoNLL-U format

```
# sent_id = ssj12.45.201
# text = Igralni oddelek One Toucha 557.
1 Igralni igralen ADJ Agpmsny Case=Nom|Definite=Def|Degree=Pos|Gender=Masc|Number=Sing 2 amod _ NER=O
2 oddelek oddelek NOUN Ncmsn Case=Nom|Gender=Masc|Number=Sing 0 root _ NER=O
3 One One PROPN Npmsn Case=Nom|Gender=Masc|Number=Sing 4 nmod _ NER=B-misc
4 Toucha Touch PROPN Npmsg Case=Gen|Gender=Masc|Number=Sing 2 nmod _ NER=I-misc
5 557 557 NUM Mdc NumForm=Digit|NumType=Card 4 nummod _ NER=I-misc|SpaceAfter=No
6 . . PUNCT Z _ 2 punct _ NER=O
```

We present the most common and established NER corpora for the South-Slavic languages in Table 2.1. The Slovene language has a newly published combined Training corpus SUK 1.0 (Arhar Holdt et al., 2022), which consists of ssj500k, ELEXIS-WSD and SentiCoref corpora. Unfortunately, only the WikiANN(Rahimi et al., 2019) automatic corpora are available for Bosnian and Macedonian languages.

Table 2.1: List of South-Slavic NER Corpora, which shows each corpus with the number of sentences, the number of tokens it contains, and its long name.

Sentences	Tokens	Long Name	
Slovene			
18106	400291	BSNLP 2017/21	(Piskorski et al., 2021)
9483	193611	ssj500k 2.3	(Krek et al., 2021)
2024	31233	ELEXIS-WSD 1.0	(Martelli et al., 2022)
18139	391526	SentiCoref 1.0	(Žitnik, 2019)
Croatian			
820	18704	BSNLP 2017/21	(Piskorski et al., 2021)
24780	504227	hr500k 1.0	(Ljubešić et al., 2018)
Serbian			
3891	86726	SETimes.SR 1.0	(Batanović et al., 2018)
Bulgarian			
18168	425438	BSNLP 2017/21	(Piskorski et al., 2021)

2.5 Cross-lingual Transfer Learning

Following the work of Prelevikj and Žitnik (2021), we wanted to explore if fine-tuning the LPMLMs with related language data improves model performance for the NER task (Ivačić et al., 2023). We examined the impact of providing additional training data on model performance for three related languages, namely Slovene, Croatian, and Serbian. We demonstrated that LPMLMs performance enhancement is most significant when the target language has a limited training corpus. Furthermore, we revealed that introducing additional language training data does not negatively affect the model’s performance for the target language. In every experiment we conducted, XLM-RoBERTa outperformed mBERT, and we also attained a state-of-the-art F1 score for the Slovene language NER model.

Recently, the GPT-derived ChatGPT model has demonstrated the potential for state-of-the-art performance in NER by employing zero-shot transfer learning. Although, most

likely, the model hasn't been trained on Slovene language NER datasets, it still manages to classify arbitrary text accurately. The ability to accurately classify even arbitrary "unseen" NER labels demonstrates the potential for new research avenues. These new approaches markedly differ from our traditional single-task supervised systems. Instead of focusing on a single problem, the model is trained across multiple languages, diverse NLP tasks, and various objectives. This potential is discussed in more detail in Chapter 4.

Chapter 3

Computational Framing Analysis

Framing is a selection of particular aspects of factual events and rendering them salient to promote a particular interpretation, evaluation, and/or solution (Entman, 1993, 2003). Alternatively, a frame was presented as “a central organizing principle that holds together and gives coherence and meaning to a diverse array of symbols” (Gamson et al., 1992).

Computational Framing Analysis (CFA) is a research field of communication framing detection and analysis using computation. Since news texts are the most common type of communication used in the research community for CFA, we will be presenting its application in this context. Although framing is casually defined, computational framing analysis has been studied diversely (Ali & Hassan, 2022).

3.1 Framing Theory and Its Relevance to News Analysis

The same facts in news articles can be presented in various ways. Media plays a crucial role in our society; therefore, observing and studying news framing is crucial to understanding its influences. Framing can have many social consequences since it can shape public discourse and induce strong emotions towards specific topics that could otherwise be overlooked or marginalized. It can promote specific perspectives influencing the public’s perception, leading to different interpretations through carefully selected narratives. It is often hidden in news articles and presented as content for casual reading. News Framing can reflect dominant group influences by repeating and reinforcing specific views and topics. Therefore Computational Framing Analysis is very important for today’s overall understanding of ever-growing news content. Computational Framing Analysis assists individuals in comprehending the reasons behind the dominance of particular viewpoints, while other equally significant perspectives have not gained prominence.

3.2 Traditional Approaches to CFA

CFA can be considered a Topic Modelling problem in an unsupervised setting or a multi-label classification problem when the supervised methods are applied. Topic modelling is a type of statistical modelling used in Natural Language Processing (NLP) and machine learning for discovering abstract “topics” that occur in a collection of documents. A topic in this context refers to a set of words that are frequently seen together in a document and capture a specific theme or subject. When supervised methods in CFA are used, the text is usually annotated with multiple predetermined labels on various levels: whole article, title, body, paragraphs, or text spans.

3.2.1 Unsupervised and Semi-supervised Approaches to CFA

Traditionally CFA utilized various unsupervised methods. Unsupervised methods do not require labelled data but aim to discover hidden patterns or structures in the text. Some common unsupervised techniques for framing analysis include Topic Modelling (TM) with Latent Dirichlet Allocation (LDA) algorithm (Blei et al., 2003). These algorithms aim to discover underlying topics or themes in large collections of documents and can be employed to reveal latent frames in the text (Blei, 2012). The basic idea is that documents are represented as random mixtures over latent topics, where the topic is a probability distribution over a fixed vocabulary. DiMaggio et al. (2013) employed LDA topic modelling to investigate the framing of government support for artists and arts organizations, with each topic being regarded as a distinct frame. Structural topic modelling (STM) was later applied in a similar fashion by Gilardi et al. (2020) and Roberts et al. (2014), which included not only topics from the text but also covariates present in document metadata (e.g. gender, origin, political affiliation, time trends etc...). The STM was further improved by Nguyen et al. (2015), who introduced the supervised Hierarchical Topic Model - Supervised Hierarchical Latent Dirichlet Allocation (SHLDA). SHLDA can discover a hierarchy of topics from a collection of documents, each associated with the ideological (liberal, conservative) score of the text author (Member of the U.S. Congress). In the learned hierarchy, first-level nodes map to the general topic (agenda, issue), while second-level nodes represent ideologically polarized frames of the corresponding issue. Walter and Ophir (2019) proposed “Analysis of Topic Model Networks” where they applied LDA topic modelling and then created a semantic network. In this network, topics represent nodes; their similarity relationships serve as edges. Lastly, a community detection algorithm was employed to group the topics into distinct communities within the network based on the prevalence of these topics in similar documents.

K-means clustering with TF-IDF vector model was applied by Burscher et al. (2016) to CFA using also mean word sentiment from SentiWord to detect a positive or negative tone for a given frame. The “elbow” method was used to predetermine the number of frames - clusters. In contrast to topic modelling, k-means clustering follows a single-membership approach, assigning each document to just one cluster.

3.2.2 Supervised Approaches to CFA

Boydstun et al. (2014), Boydstun et al. (2013) developed a manually annotated corpus on which they used logistic regression binary text classifiers. Card et al. (2016) used logistic regression classifier with unsupervised Dirichlet Persona Model (Bamman et al., 2013) (DPM). The logistic regression classifier with DPM achieved improved performance when utilizing a wide range of NLP features (e.g. unigrams, bigrams, parts of speech, named entities, dependency tuples, ...) compared to the simple bag-of-words approach.

From this point forward, neural network models have been CFA’s dominant supervised learning approaches. Naderi and Hirst (2017) out-performed LDA and TF-IDF approaches on sentence-level classification tasks by employing bidirectional LSTM (Graves et al., 2013; Hochreiter & Schmidhuber, 1997) (Bi-LSTM) and gated recurrent units (Cho et al., 2014) (GRU). Ji and Smith (2017) out-performed logistic regression classifier (Card et al., 2016) using Bi-LSTM with discourse dependency tree embedding, where the attention mechanism was used for composing child nodes representation. Both Bi-LSTM methods attained optimal results when using pre-trained GloVe (Pennington et al., 2014) as the source for input word embeddings.

S. Liu et al. (2019) was one of the first in testing transformer-based Large Pre-trained Models (LPMs), showing that BERT (Devlin et al., 2018) surpassed Bi-LSTM and GRU

models equipped with attention mechanisms. Khanehazar et al. (2019) demonstrated that LPMs also significantly out-perform support vector machines (SVM) in all cross-context (across different corpora) and cross-subject (immigration and same-sex marriage) CFA tasks. It was again evident that RoBERTa (Y. Liu et al., 2019) out-performs BERT in framing classification tasks and can be further improved with Multi-task learning where parameters are shared for the first eleven transformer layers (Huguet Cabot et al., 2020). Mendelsohn et al. (2021) demonstrated simultaneous classification for multiple framing typologies using RoBERTa on custom social media corpus. Apart from Field et al. (2018), which used a clustering lexicon construction approach with an automatic translation to the Russian language, multilingual CFA wasn't widely studied.

3.3 Recent Research of Multilingual CFA

Multilingual CFA recently gained traction with SemEval-2023 task 3, where sub-task two specifically targeted news framing in multiple languages (Piskorski et al., 2023). The provided corpora for the task contained only a few hundred samples for English, French, German, Italian, Polish, and Russian languages. At the same time, no samples were available for Spanish, Greek, and Georgian. So the task was not only multilingual but also a zero- and a few-shoot supervised learning task.

The winning solution (Liao et al., 2023) for four languages, French, German, Italian, and Polish, used the XLM-RoBERTa-Large transformer model as an encoder and implemented a multilingual and multi-label contrastive model on top of it. Their system came second in the remaining two languages with training data (English and Russian). Contrastive learning has been a widespread technique in computer vision for a long time. The process involves selecting a random example as an anchor, followed by choosing positive and negative examples in relation to the anchor. Subsequently, positive examples are drawn closer to the anchor, while negative examples are pushed further away. They calculated combined loss from binary cross entropy loss and contrastive loss obtained using a supervised architecture (Khosla et al., 2021) adapted to a multi-label problem. Following the work on SimCSE model (Gao et al., 2022), they produced two independent views of the data obtained from the encoder using a dropout and then used one sample for contrastive loss and the other for binary cross entropy loss calculation.

One of the solutions (Reiter-Haas et al., 2023) that won a challenge on an "unseen" Spanish language also used contrastive learning with a transformer model. Reiter-Haas et al. (2023) used a two-phase multi-stage training procedure which fine-tunes sentence embeddings in a contrastive manner inspired by SETFIT (Tunstall et al., 2022). The model is built on top of sentence transformers (Reimers & Gurevych, 2019) encoder where they used a contrastive loss function modelled after the HeroCon-Loss (Zheng et al., 2022). The initial phase is comprised of head pre-training and contrastive fine-tuning stages. The result of the two-stage first phase can be used for zero-shot prediction. Extended first phase with post-training and later weight reinitialization for additional second phase fine-tuning was used for a few-shot-based prediction.

3.4 Datasets and Resources

Boydstun et al. (2014), Boydstun et al. (2013) introduced a codebook referred to as the "Policy Frames Codebook" (PFC) and developed a manually annotated corpus. The codebook was developed through brainstorming and iterative application to randomly selected texts based on the framing definition of Entman (1993). Through this process, they established 15 framing dimensions shown in Table 4.1.

Table 3.1: List of frame dimensions defined by Boydstun et al. (2014), Boydstun et al. (2013) and used in MFC corpus.

Economic: costs, benefits, or other financial implications
Capacity and resources: availability of physical, human or financial resources, and capacity of current systems
Morality: religious or ethical implications
Fairness and equality: balance or distribution of rights, responsibilities, and resources
Legality, constitutionality and jurisprudence: rights, freedoms, and authority of individuals, corporations, and the government
Policy prescription and evaluation: discussion of specific policies aimed at addressing problems
Crime and punishment: effectiveness and implications of laws and their enforcement
Security and defence: threats to the welfare of the individual, community, or nation
Health and safety: health care, sanitation, public safety
Quality of life: threats and opportunities for the individual’s wealth, happiness, and well-being
Cultural identity: traditions, customs, or values of a social group in relation to a policy issue
Public opinion: attitudes and opinions of the general public, including polling and demographics
Political: considerations related to politics and politicians, including lobbying, elections, and attempts to sway voters
External regulation and reputation: international reputation or foreign policy of the U.S.
Other: any coherent group of frames not covered by the above categories

Utilizing the PFC, Card et al. (2015) provided a manually-annotated dataset of news reports called the “Media Frames Corpus” (MFC). MFC is the most commonly used dataset for supervised CFA approaches. It comprises 13 national U.S. newspaper articles from 1990 to 2012, covering three policy issues: immigration, smoking, and same-sex marriage. Multiple coders annotated each headline, article lead, article body text spans and overall framing of an article.

Another annotated dataset, the “Gun Violence Frame Corpus” (S. Liu et al., 2019), was utilized in neural network-based models explored earlier and is also based on the framing definition of Entman (1993). In this dataset, the authors manually annotated 1,300 news headlines gathered from 21 U.S. media outlets. Employing nine pre-defined codes derived from the literature, multiple coders annotated each headline.

Lind et al. (2020) provides automatically and manually annotated news article corpus in seven languages. This corpus contained five framing dimensions (Economy, Labour market, Welfare, Security, and Culture) regarding migration issues and was also the basis for the Slovene language corpus we started to develop. The corpus includes annotations for both the framing of the title and the overall framing of the article.

id	country	source_id	source	published	title	body	migration	fr_cul	fr_eco	fr_lab	fr_sec	fr_wel
2015121907162	SI	270	Vecer.com	2015-12-18T23:4	Kratke	Kratke	1	0	0	0	1	1
2015120919100	SI	240	Rtvslo.si	2015-12-09T18:1	V nevarnosti tako schengen kot tudi pri	V nevarnosti tako	1	0	0	0	1	1
2016011217500	SI	290	Zurnal24.si	2016-01-12T17:0	Bo Slovenija zavračala hrvaške vlake? Ker Avstrija upo	Ker Avstrija upo	1	0	0	0	1	1
2015092111593	SI	240	Rtvslo.si	2015-09-21T11:1	Juncker obvladovanje razmer s prebeži Juncker obvlado	Juncker obvlado	1	0	0	0	0	1
20160102211111	SI	260	Slovenskenovice	2016-01-02T10:1	Brglez: Ograja je neuspeh Slovenije in	PREDSEDNIK D	1	0	0	0	1	0
2016011915323	SI	60	Dolenjskolist.si	2016-01-18T20:4	Zavrnili pobudo za odstranitev žičnate	Zavrnili pobudo	1	1	0	0	1	1
2016010616021	SI	30	Delo.si	2016-01-06T15:4	Schengen pomemben, a nadzor ostaja V Bruslju so se z	V Bruslju so se z	1	0	0	0	1	1
2015102613503	SI	40	Demokracija.si	2015-10-26T13:0	Bruselj: Slovenija zaprosila za približno	Napisal STA	1	0	0	0	0	1
2015091805594	SI	130	Mladina.si	2015-09-17T22:4	Koalicija sebičnih	Koalicija sebičnih	1	1	0	0	1	0
2016011409554	SI	230	Reporter.si	2016-01-14T06:1	Osem polovic svinskih glav na gradbišču	Osem polovic sv	0	0	0	0	0	0
2016012010253	SI	270	Vecer.com	2016-01-19T23:4	Nizozemski premier o beguncih: Zmanj Nizozemski pren	Zmanj Nizozemski pren	1	0	1	0	1	0
2016033019152	SI	200	Primorske.si	2016-03-30T11:4	V Nemčiji dvigajo prah načrti vlade o se	V Nemčiji dvigaj	1	1	0	1	1	1

Figure 3.1: REMINDER project framing corpus (Lind et al., 2020) structure example in the Slovene language with five framing labels (Culture, Economy, Labour market, Security, and Welfare) in the right-most columns and each article in a row.

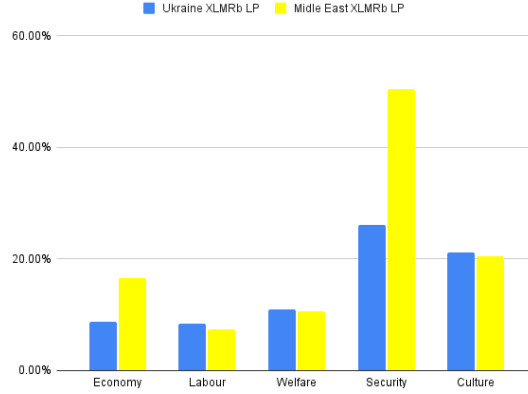


Figure 3.2: Preliminary research results comparing two EU migration crises using the Slovenian corpora predictions with a Zero-shot technique, showing the relative differences in Economy and Security news framing.

3.5 Our Preliminary Research

Our task was to compare Slovenian news frames of two European migration crises caused by the war in Syria and Ukraine. We used an automatically annotated news corpus (Lind et al., 2020) developed for the REMINDER project to train the multilingual frame classifiers. Corpus consisted of migration news articles in seven languages and was annotated with five universal frames regarding migration. We aimed to test Zero-shot Framing detection for the Slovene language not included in the corpus. Therefore, we have chosen two multilingual pre-trained transformer models XLM-RoBERTa and BERT and fine-tuned them on the migration corpus for multi-label classification. Furthermore, we used two classical transformation methods to tackle multi-label classification problems: Binary Relevance and Label Power-set (Ganda & Buch, 2018). Further on, we constructed the Slovene migration corpus for the two periods, used the fine-tuned models for news framing prediction, and compared the results. The initial findings support our hypothesis that news framing in Slovenia differed between the two periods; however, additional research is required to substantiate this conclusion further.

Chapter 4

Trends and Challenges

This chapter provides a concise overview of key considerations for both Named Entity Recognition (NER) and Computational Framing Analysis (CFA), followed by potential improvements to our previous research. The chapter concludes by outlining further areas of research in news analysis applied solutions that we intend to investigate.

4.1 Named Entity Recognition

4.1.1 General Considerations

We argue that the traditional CoNLL NER annotations, which have been widely adopted, contain insufficient labels considering the capabilities of today’s Large Pre-trained Multilingual Models (LPMMs). While this has been recently addressed in Malmasi et al. (2022), there is still a challenge for minority and low-resource languages. A related challenge we see is custom on-the-fly labelling instead of pre-selected static one, where unseen labels by the training are given at inference time. This can be addressed with more research on Zero-shot learning techniques regarding unseen label discovery.

4.1.2 Past Research Improvements

Since the short paper submission (Ivačič et al., 2023), we have already further investigated if the model is affected by fine-tuning with broader family languages. The first results confirm that model performance is stable even if fine-tuned with 11 additional languages. Further cross-validation is necessary to validate and claim state-of-the-art results, especially considering that no significant improvements have been made to LPMMs.

4.2 Computational Framing Analysis

4.2.1 General Considerations

As most recent research in CFA has shifted towards supervised learning with LPMMs, we identify the primary challenges to further advancements in the utilization of the same Policy Frames Codebook (PFC) as a foundation for the most widely used corpora: “Media Frames Corpus” (MFC) and “Gun Violence Frame Corpus” (GVFC). The labels derived from the PFC are often misinterpreted as an additional layer of topics, referring to the content presented in the news rather than addressing the fundamental framing question of how the news is presented (Ali & Hassan, 2022). Despite the vague definition of framing, the PFC labels remain static, which may be disputable when alternative news framing emerges in response to a new issue or agenda. Following the research initiated by SemEval-2023

task 3 (Piskorski et al., 2023), we believe the general trends go in the right direction. The trend of limited-size corpora encourages the pursuit of further advancements in zero- and few-shot learning approaches, which again reduces the need for large, manually annotated corpora. More diverse corpora with a broader aim to develop models that could further generalize the framing concept without solely counting on the PFC codebook. In this context, the latest advancements in text-to-text transformers could offer potential benefits by generalizing the framing concept beyond predefined labels, allowing for more flexible generalizations derived from these labels. Therefore this will be the focus of our future research.

4.2.2 Past Research Improvements

Since only initial work has been conducted regarding CFA and the Slovene language, there are many, even essential, improvements to be made. Apart from more rigorous cross-validation with removing unnecessary problem transformation methods, we should finish the started manual corpora annotation with multiple coders to achieve mutual annotator agreement. Next, we could repeat the recent and relevant SemEval-2023 experiments to begin the endeavour to research framing concept generalization with few-shot learning.

4.3 Additional Areas for News Analysis Applied Solutions

In addition to past NER and initial CFA research, we intend to broaden the area of advanced applied solutions to news analysis through follow-up studies.

4.3.1 Extreme Multi-label Hierarchical Classification

There is also a potential to improve general multi-label text classification by leveraging prior and additional CFA work. The research to develop concrete applied solutions for media monitoring, capable of extreme multi-label hierarchical classification handling of >1,000 topics, led to tools for automatically classifying documents into industry-standard IPTC categories across languages. Another challenge lies in addressing the automatic classification of a ten times larger number of customized, client-tailored topics.

4.3.2 Multi-level Online Clustering

Media monitoring needs to be performed in real-time: grouping articles by their content, adding several categories of meta-information, performing analyses, and reporting from the perspective of a particular client before distributing the content. Grouping articles by their content on a small daily scale can be achieved efficiently with an appropriate document representation and additional meta-information. The challenge emerges when a news archive expands to millions of documents, necessitating the pre-calculation of various clustering levels to accommodate the constant influx of new articles. Various approaches have been suggested (Jiang et al., 2021; Miranda et al., 2018; Santos et al., 2022), but the subject remains an open research question.

4.3.3 Extractive and Abstractive Summarisation

Methods for media monitoring tasks must be capable of summarizing single documents and multiple news articles or streams of social media posts and producing output specific to a given client. An open research question is how to handle these different summarisation scenarios using extractive and abstractive summarisation methods. Extractive models select

significant sentences and must do this across different media and different languages. Abstractive models must then generate text summaries from these sentences in less-resourced languages, e.g. Slovene. From the applied perspective, summarisation is currently manually performed or using a very simplistic extraction of words surrounding the target. Significant advancements have been made with T5 (Raffel et al., 2020), GPT-3 (Brown et al., 2020), PaLM (Chowdhery et al., 2022), LLaMa (Touvron et al., 2023) models and summarisation, but “model hallucination” still poses a challenge, especially where factual information presented is essential.

Table 4.1: Example state-of-the-art GPT derivative ChatGPT (GPT-4) model hallucinations.

	I want you to act as a natural language processing scientist. I will provide you with details about the topic. You should be knowledgeable of natural language processing terminology and metrics. You will answer as a scientist and use only facts. My first request is: can you write about some examples of news framing studies but skip writing descriptions of news framing and write more about large pre-trained multilingual models like XLM-RoBERTa or multilingual BERT and how they are used.
Q:	Try not to make an introduction and also include scientific papers.
	...XLM-RoBERTa, proposed by Conneau et al. (2020), is another large pre-trained multilingual model. It builds upon RoBERTa (Liu et al., 2019) and is pre-trained on a corpus of 100 languages. This model has been successfully used in news framing studies,
A:	such as the work of Kardas et al. (2021), which explored the framing of the COVID-19 pandemic in Polish news. By ...
Q:	Tell me which paper was written by Kardas et al. (2021), which explored the framing of the COVID-19 pandemic in Polish news.
	I apologize for the confusion in my previous response. Upon further review, I could not find a paper by Kardas et al. (2021) that explored the framing of the COVID-19 pandemic in Polish news. The information provided in my earlier response was incorrect,
A:	and I apologize for any confusion this may have caused.

4.4 Summary

We presented two traditional news analyses NLP tasks with our ongoing research. The NLP research community increasingly gravitates towards multilingual models, utilizing Few-shot and Zero-shot learning techniques. In this context, Few- and Zero-shot learning refers to the learner handling not only “unseen” training data but also “unseen” tasks. Advancements in this field significantly benefit low-resource languages, which often suffer from a scarcity of training data. The great research potential lies in large language, text-to-text transformer models trained on many languages and tasks, with the ability to solve arbitrary NLP problems. Leveraging these emerging paradigms, we plan to continue our research focusing on applied solutions for news analysis.

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